

LESSON 126 (0.5) GROUND and (0.8) SOLO

Lesson Objective

The purpose of this lesson is to continue to enhance the applicant's confidence and abilities to operate the aircraft during solo flight. This lesson will allow the applicant time to practice go-around or rejected landing procedures, normal and crosswind takeoff, and landings as well as traffic pattern procedures.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

References

- Airplane Flying Handbook. (FAA-H-8083-3C)
 - Chapter 06 - Takeoffs and Departure Climbs.
 - Chapter 08 - Airport Traffic Patterns.
 - Chapter 09 - Approaches and Landings.
- Pilot's Handbook of Aeronautical Knowledge. (FAA-H-8083-25B)
 - Chapter 14 - Airport Operations.

Instructor Guidance

- This is the student's first unsupervised solo lesson — there is no dual portion preceding it. Your pre-brief and go/no-go assessment before releasing the student are therefore especially critical. Take the time to thoroughly evaluate weather conditions, the student's readiness, and their awareness of the plan of action before signing off.
- Before the student departs, quiz them specifically on the questions listed in the unsupervised solo plan of action — expected runway, wind direction and correction technique, anticipated ATC instructions after departure, and go-around trigger scenarios. If the student struggles to answer these confidently, delay the solo and address the gaps.
- Verify the student's solo endorsement currency before every unsupervised solo flight. Per 14 CFR §61.87, endorsements are valid for 90 calendar days. If expired or close to expiring, re-endorse before the student departs.
- Track the student's solo flight using FlightRadar24 or FlightAware for the duration of the flight. Confirm that all required line items are being completed and that the student remains in the expected area of operations.
- Remind the student before departure that all landings must be full-stop taxi-backs — no touch-and-go landings. This is a non-negotiable FIT requirement for solo operations and must be briefed explicitly every time.
- Monitor the student's radio calls via the handheld portable radio throughout the solo. If communications are not audible, do not release the student until a solution is found.
- Remind the student to announce "student pilot solo" on initial contact with ATC. This is a safety practice that ensures tower is aware of the student's status.
- This lesson is closed out by dispatch — make sure the student understands this process and

knows what is expected of them upon returning. Ensure the student completes their logbook entry independently, including an accurate description of the flight activities.

- Follow up with the student after the solo flight to address any questions, concerns, or debrief items. Even though this is an unsupervised solo, the post-flight debrief is still a valuable teaching opportunity. Ask the student how they felt about their go-around decision-making specifically — this is one of the highest-risk areas in solo pattern operations.

Lesson Content (Ground)

Unsupervised solo plan of action

- Normal and crosswind takeoffs
 - Ask the student:
 - What runway are you expecting to take off from?
 - What direction is the wind coming from and how will you apply correction as a result?
- Traffic pattern procedures - Towered
 - Ask the Student
 - What instructions are you expecting after departure?
- Normal and crosswind landings
 - Ask the student:
 - What direction is the wind coming from and how will you apply correction as a result?
- Go-around/rejected landing
 - Ask the student:
 - What situation might cause a go-around?
 - What is the procedure for going around?
 - Perform a go-around if necessary.
 - Unstable approach.
 - Vehicle, wildlife, etc. on the runway.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Solo)**Checklist Procedures**

- Student pilot solo

Engine Starting

- Student pilot solo

Operation of the Communications and Navigation Units

- Student pilot solo

Radio Communication Procedures and ATC Phraseology

- Student pilot solo

Visual Scanning and Collision Avoidance

- Student pilot solo

Taxiing

- Student pilot solo

Normal and Crosswind Takeoff

- Student pilot solo

Traffic Patterns

- Student pilot solo

Go-Arounds or Rejected Landings

- Student pilot solo

Normal and Crosswind Landings

- Student pilot solo

Postflight Procedures

- Student pilot solo.
- Ring the bell!
 - "Let it be known to all on this day, (day,month,year), Mr./Mrs xxx, has demonstrated the knowledge, skills, good judgement and mastery of the Piper Warrior to complete his/her first solo".

Post-Brief

- Student self-critique.
- Lesson grading.
 - With Primary instructor

- Logbook completion.
 - Description of activities input by student.
- Schedule the next lesson.
- Preview the next lesson → 124.
- Assign homework:
 - Read chapter 06 of the Airplane Flying Handbook - Takeoffs and Departure Climbs:
 - Short field takeoffs.
 - Soft field takeoffs.
 - Read chapter 08 of the Airplane Flying Handbook - Airport Traffic Patterns:
 - Short field landings.
 - Soft field landings.
 - Read chapter 05 of the Pilot's Handbook of Aeronautical Knowledge - Aerodynamics of Flight
 - Wingtip Vortices.

Completion Standards

This lesson is complete when the applicant can:

1. Execute a local solo in accordance with all FTA, FAA and local procedures and regulations.